

ASSESSMENT TOOL 1 – Analytical Problem Solving Questions

Course Code:	CSA0315	Course Title:	Data Structures
CO Assessed:	CO2	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	ADT array, Operations on arrays, Searching Algorithms, Linear Search, Binary Search, Suffix Arrays & LCP Arrays ADT, Linked List, Stack, Queue, Applications.		
Student Name:	K.B.Tharunika	Reg. No.:	192525276
Section / Batch:	2025	Date:	

Code of Conduct

I K.B.Tharunika (192525276)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.B.Tharunika

Instructions

- This test contains 3 Analytical Problem Solving questions. Total: 100 Marks.
- When a student answer using a analytical problem, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are not permitted in the class. Take it as test and submit in the class.

Section / Topic	Focus	Q Nos	Marks
Stack Data Structures	Stack – top movements	1	30
Queue Data Structures	Circular Queue	2	40
Application of Stack	Infix to post-fix	3	40
GRAND TOTAL:			100 Marks



Name: K. Tharunika

Regno: 192525276

Course Name & course Data Structures

Code: CSE-0315

Department: B.Tech AI EML

Date: 28/07/26

Co-2 AT-1 Test
Analytical problem
Solving

26/36

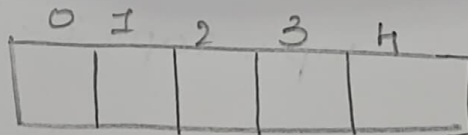
1.

Perform the following operation in LIFO data structure having an array size of 5, push(45), push(22), push(11), push(77), pop(), push(44), push(33), pop(), pop(), pop() print the peak element of data structure and its index position with a neat sketch explain in detail

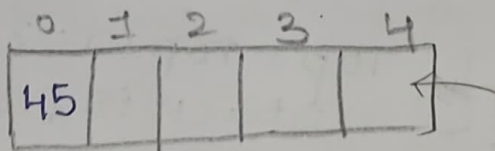
Sol.

Given, size of stack = 5

Initial state Top = -1



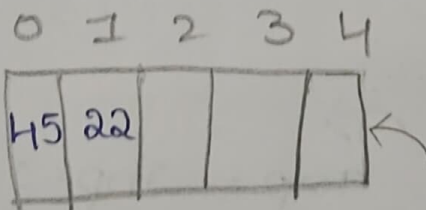
Push(45)



Top = Top + 1

$$= -1 + 1 = 0$$

push(22)

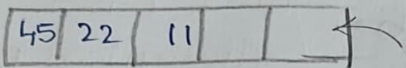


top = Top + 1

$$= 0 + 1 = 1$$

push(11)

0 1 2 3 4

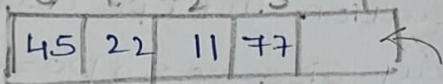


$$\text{top} = \text{top} + 1$$

$$= 1 + 1 = 2$$

push(77)

0 1 2 3 4

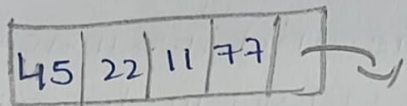


$$\text{top} = \text{top} + 1$$

$$= 2 + 1 = 3$$

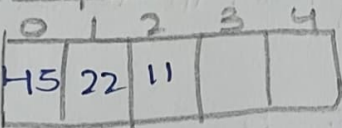
pop()

0 1 2 3 4



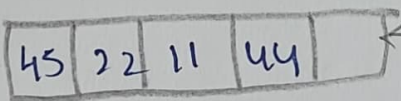
$$\text{top} = \text{top} - 1$$

$$= 3 - 1 = 2$$



push(44)

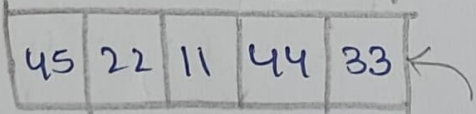
0 1 2 3 4



$$\text{top} = \text{top} + 1 = 2 + 1 = 3$$

push(33)

0 1 2 3 4

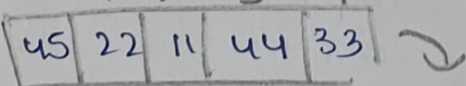


$$\text{top} = \text{top} + 1 = 3 + 1 = 4$$

Stack is full

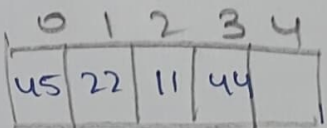
pop()

0 1 2 3 4



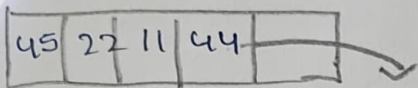
$$\text{top} = \text{top} - 1 = 4 - 1$$

$$= 3$$

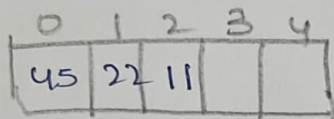


Pop()

1 2 3 4

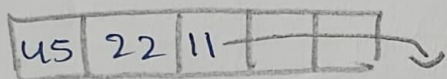


$$\text{Top} = \text{Top} - 1 = 3 - 1 = 2$$

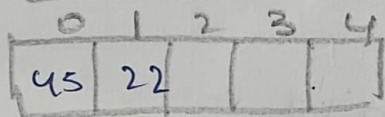


Pop()

0 1 2 3 4



$$\text{Top} = \text{Top} - 1 = 2 - 1 = 1$$



∴ The peak element of the data structure is "22" and the index of data structure is "1".

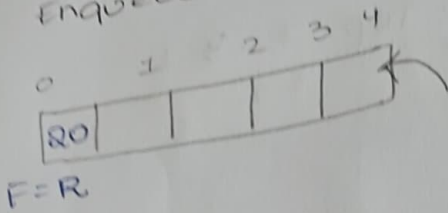
2. Perform the following operation in circular Queue having a size of 5 Enqueue(20), Enqueue Enqueue(30), Dequeue(), Enqueue(80), Dequeue Enqueue(90), enqueue(100), Dequeue(), Dequeue Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions, discuss in detail.

Given, Size of Queue = 5

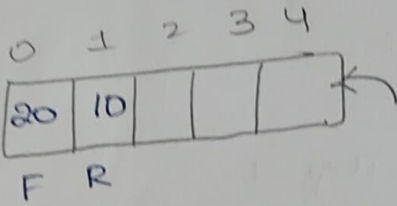
Initial state Rear = Front = -1

$$\text{front} = \text{Rear} = 0$$

Enqueue (20)

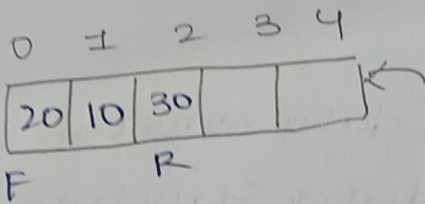


Enqueue (10)



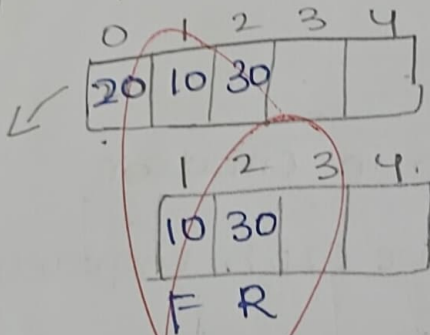
$$\begin{aligned} \text{Rear} &= \text{Rear} + 1 \\ &= 0 + 1 = 1 \end{aligned}$$

Enqueue (30)



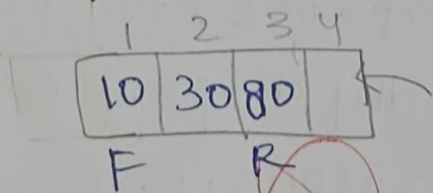
$$\begin{aligned} \text{Rear} &= \text{Rear} + 1 \\ &= 1 + 1 = 2 \end{aligned}$$

Dequeue ()



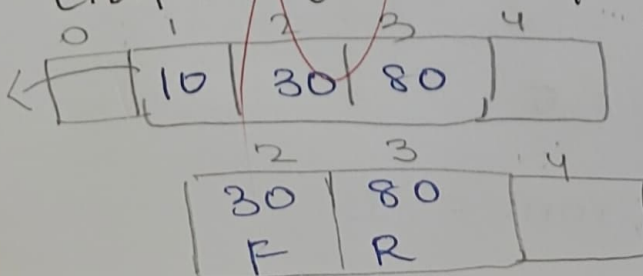
$$\begin{aligned} \text{Front} &= \text{Front} + 1 \\ &= 0 + 1 = 1 \end{aligned}$$

Enqueue (80)



$$\begin{aligned} \text{Rear} &= \text{Rear} + 1 \\ &= 2 + 1 = 3 \end{aligned}$$

~~Enqueue (90)~~ Dequeue ()



$$\begin{aligned} \text{Front} &= \text{Front} + 1 \\ &= 1 + 1 = 2 \end{aligned}$$

queue (90)

2	3	4
30	80	90
F		R

$$\text{Rear} = \text{Rear} + 1 \\ = 2 + 1 = 3$$

enqueue (100)

0	1	2	3	4
100		30	80	90
R		F		

$$\text{Rear} = \text{Rear} + 1 \\ = 4 + 1 = 0$$

dequeue ()

0	1	2	3	4
100		30	80	90

0	1	3	4
100		80	90
R		F	

$$\text{Front} = \text{Front} + 1 \\ = 2 + 1 = 3$$

dequeue ()

0	1	3	4
100		80	90

0	1	4
100		90
R		F

$$\text{Front} = \text{Front} + 1 \\ = 3 + 1 = 4$$

enqueue (60)

0	1	2	3	4
100	60			90
F				R

$$\text{Front} = 4$$

$$\text{Rear} = \text{Rear} + 1 \\ = 0 + 1 = 1$$

enqueue (90)

0	1	2	3	4
100	60	90		90

$$\text{Front} = 4$$

$$\text{Rear} = \text{Rear} + 1 \\ = 1 + 1 = 2$$

Front element is at position '4' with value 10
 Rear element is at position '2' with value 1

3. Convert the following infix expression into a post fix expression

$$B - C C / D + E \cdot 1 \cdot F * G) / H) * J$$

Also evaluate the given post fix expression

a.) $9 \ 3 \ 8 \ 4 \ / \ * \ +$

b.) $6 \ 2 \ + \ 6 \ 3 \ / \ *$

Scanned Char	Stack content	Post fix expression
B	B	B
-	- B	B
C	- C	B C
C	- C C	B C C
/	- C C /	B C C /
D	- C C / D	B C C / D
+	- C C / D +	B C C / D +
C	- C C / D + C	B C C / D + C

Char	Stack content	Post fix expression						
	<table border="1"><tr><td>+</td><td>C</td><td>+</td><td>C</td></tr></table>	+	C	+	C	BCD/E		
+	C	+	C					
	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>C</td><td>.</td><td>1</td></tr></table>	-	C	+	C	.	1	BCD/LE
-	C	+	C	.	1			
F	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>C</td><td>.</td><td>1</td></tr></table>	-	C	+	C	.	1	BCD/LEF
-	C	+	C	.	1			
*	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>C</td><td>*</td></tr></table>	-	C	+	C	*	BCD/LEF.*	
-	C	+	C	*				
G	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>C</td><td>*</td></tr></table>	-	C	+	C	*	BCD/LEF.*G	
-	C	+	C	*				
)	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>C</td><td>*</td></tr></table>	-	C	+	C	*	BCD/LEF.*G	
-	C	+	C	*				
/	<table border="1"><tr><td>-</td><td>C</td><td>+</td></tr></table>	-	C	+	BCD/LEF.*G			
-	C	+						
H	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>/</td></tr></table>	-	C	+	/	BCD/LEF.*G*		
-	C	+	/					
)	<table border="1"><tr><td>-</td><td>C</td><td>+</td><td>/</td></tr></table>	-	C	+	/	BCD/LEF.*G*		
-	C	+	/					
*	<table border="1"><tr><td>-</td><td></td></tr></table>	-		BCD/LEF.*G*				
-								
)	<table border="1"><tr><td>-</td><td>*</td></tr></table>	-	*	BCD/LEF.*G*				
-	*							
End	Empty	BCD/LEF.*G*						

a.) 9384 / * +

9	<table><tr><td>9</td><td></td></tr></table>	9				
9						
3	<table><tr><td>9</td><td>3</td><td></td></tr></table>	9	3			
9	3					
4	<table><tr><td>9</td><td>3</td><td>4</td><td></td></tr></table>	9	3	4		
9	3	4				
8	<table><tr><td>9</td><td>3</td><td>4</td><td>8</td><td></td></tr></table>	9	3	4	8	
9	3	4	8			

$$/ \quad \boxed{9 \mid 3 \mid 2}$$

$$* \quad \boxed{9 \mid 6 \mid}$$

$$+ \quad \boxed{15}$$

output = 15 "

$$8/4 = 2$$

$$3 \times 2 = 6$$

$$9 + 6 = 15$$

b.) $6 \ 2 + 6 \ 3 \ 1 \ *$

$$6 \quad \boxed{6 \mid}$$

$$2 \quad \boxed{6 \mid 2 \mid}$$

$$+ \quad \boxed{8 \mid}$$

$$6 \quad \boxed{8 \mid 6 \mid}$$

$$3 \quad \boxed{8 \mid 6 \mid 3 \mid}$$

$$/ \quad \boxed{8 \mid 2 \mid}$$

$$* \quad \boxed{16 \mid}$$

output = 16 "

Annexure A – Analytical Problem Solving

1) Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the peak element of the data structure and its index position. With a neat sketch, explain in detail.

2) Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.

3) Convert the following infix expression into a post fix expression.

$B - (C/D + (E \% F * G) / H) * J$.

Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 /*

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge

Course Coordinator

Program Director

Signature: _____

Signature: _____

Signature: _____

Date: _____

Date: _____

Date: _____